Ligjerata 7a

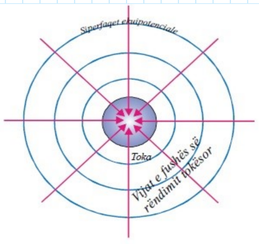
Ligji i Njutonit për gravitetin

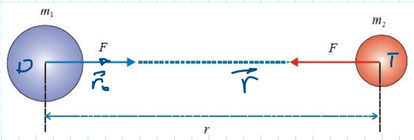
$$F = \gamma \frac{m_1 m_2}{r^2}$$

$$\vec{F} = -\gamma \frac{m_1 m_2}{r^2} \vec{r}_0 \qquad \vec{r}_0 = \frac{\vec{r}}{r}$$

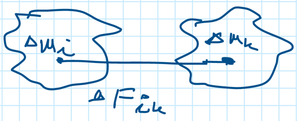
$$\vec{F} = -\gamma \frac{m_1 m_2}{r^3} \vec{r}$$

$$\boxed{\gamma = 6.67 \cdot 10^{-11} \frac{Nm^2}{kg^2}}$$





$$\boxed{\Delta F_{ik} = \gamma \frac{\Delta m_i \Delta m_k}{r_{ik}^2}}$$



$$F = \sum_{i,k} \Delta F_{ik}$$

$$\boxed{F = ma}$$

$$F = \gamma \frac{m_1 m_2}{r^2}, \quad m_1 = m, \quad m_2 = M_T, \quad r = R_T$$

$$\boxed{F = \gamma \frac{m M_T}{R_T^2}}$$

$$ma = \gamma \frac{m M_T}{R_T^2}$$

$$a = \gamma \frac{M_T}{R_T^2} = g^0$$

$$F = \gamma m g^0 = \gamma \frac{m M_T}{R_T^2}$$

$$\boxed{g^0 = \gamma \frac{M_T}{R_T^2}}$$

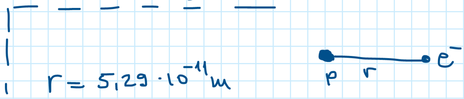
$$\boxed{g^0 = 9{,}80665 \frac{m}{s^2}}$$

$$\boxed{a = g^0}$$

Ligji i Kulonit $F = k \frac{|q_1||q_2|}{r^2}$

Ligji i Njutonit për gravitetin $F = \gamma \frac{m_1 m_2}{r^2}$

$F_g > 0$



$r = 5{,}29 \cdot 10^{-11} m$

$$F_g = \gamma \frac{m_e \cdot m_p}{r^2}$$

$$\gamma = 6.67 \cdot 10^{-11} \frac{Nm^2}{kg^2}$$

$$m_e = 9{,}11 \cdot 10^{-31} kg$$

$$m_p = 1.63 \cdot 10^{-27} kg$$

$$F_g = 6{,}67 \cdot 10^{-11} \frac{Nm^2}{kg^2} \cdot \frac{9.11 \cdot 10^{-31} kg \cdot 1.63 \cdot 10^{-27} kg}{(5{,}29 \cdot 10^{-11} m)^2} = \frac{99{,}044 \cdot 10^{-69}}{27{,}98 \cdot 10^{-22}} kg^2 \frac{Nm^2}{kg^2}$$

$$\boxed{F_g = 3{,}63 \cdot 10^{-47} N}$$

$$F_e = k \frac{|q_1| \cdot |q_2|}{r^2}$$

$$k = 8{,}99 \cdot 10^9 N \frac{m^2}{C^2}$$

$$q_1 = |-1{,}6 \cdot 10^{-19} C|$$

$$q_2 = |1{,}6 \cdot 10^{-19} C|$$

$$F_e = 8{,}99 \cdot 10^9 \frac{Nm^2}{C^2} \cdot \frac{|-1.6 \cdot 10^{-19} C| \cdot |1{,}6 \cdot 10^{-19} C|}{(5{,}29 \cdot 10^{-11} m)^2}$$

$$\boxed{F_e \approx 8{,}22 \cdot 10^{-8} N}$$

$$\frac{F_e}{F_g} = \frac{8{,}22 \cdot 10^{-8} N}{3{,}63 \cdot 10^{-47} N} = 2{,}26 \cdot 10^{39}$$

$$\boxed{F_e \gg F_g}$$